

Strategic AI Agent Development for Texas Hold'em Using Expectiminimax and Monte Carlo Simulation

CSE-440
Group:07

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Group-07 (Saba , Ema , Sumaya , Fahim)

Introducing Texas Hold'em

- Texas Hold'em is a popular poker game with hidden information
- Players receive 2 private cards and share 5 community cards
- Goal: make the best 5-card hand or force opponents to fold
- Decisions involve uncertainty, probability, and strategy
- Our project builds an AI agent to play Texas Hold'em intelligently



Project Objective



**Design a strategic
poker-playing AI**



**Handle incomplete
information**



**Make decisions
under random
card dealing**



**Simulate real
human-like poker
reasoning**



**Choose optimal
actions: Fold,
Check, Call, Raise**





Core Methods Used



Monte Carlo Hand Strength Simulation



Balanced Heuristic Evaluation



Expectiminimax Decision Algorithm



Probability + Strategy + Lookahead combined



**Estimates win probability
under uncertainty**



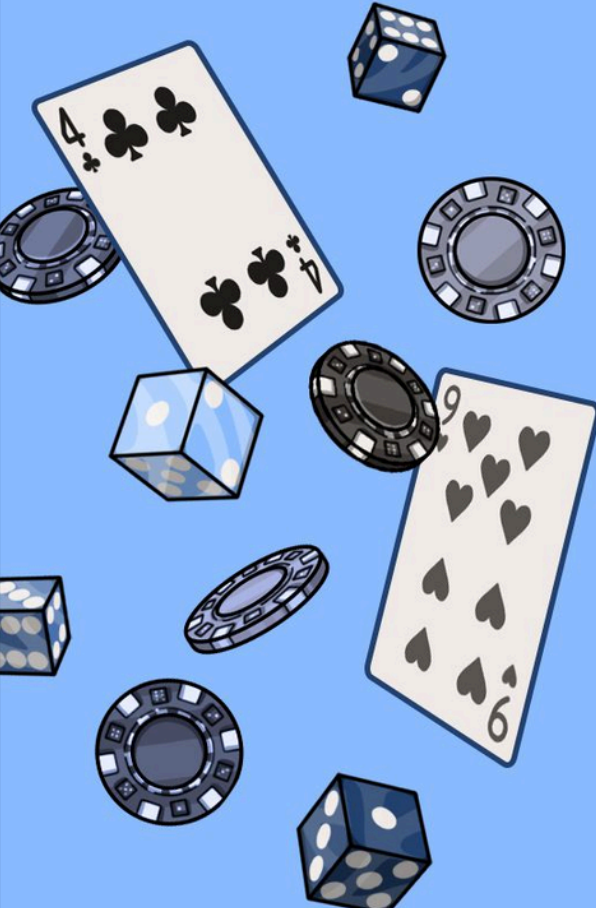
- Randomly simulates:
Opponent cards
- Remaining community
cards



**Runs hundreds to
thousands of
simulations**



**Produces realistic hand
strength values. Solves
the problem of unknown
future states**



Monte Carlo Simulation

[män-tē 'kär-'lō sim-yə-'lā-shən]

A model used to predict the probability of a variety of outcomes when the potential for random variables is present.

 Investopedia

Monte Carlo Simulation – Significance



Converts win probability into a state score



Encourages:

- Aggressive play for strong hands
- Safe folding for weak hands

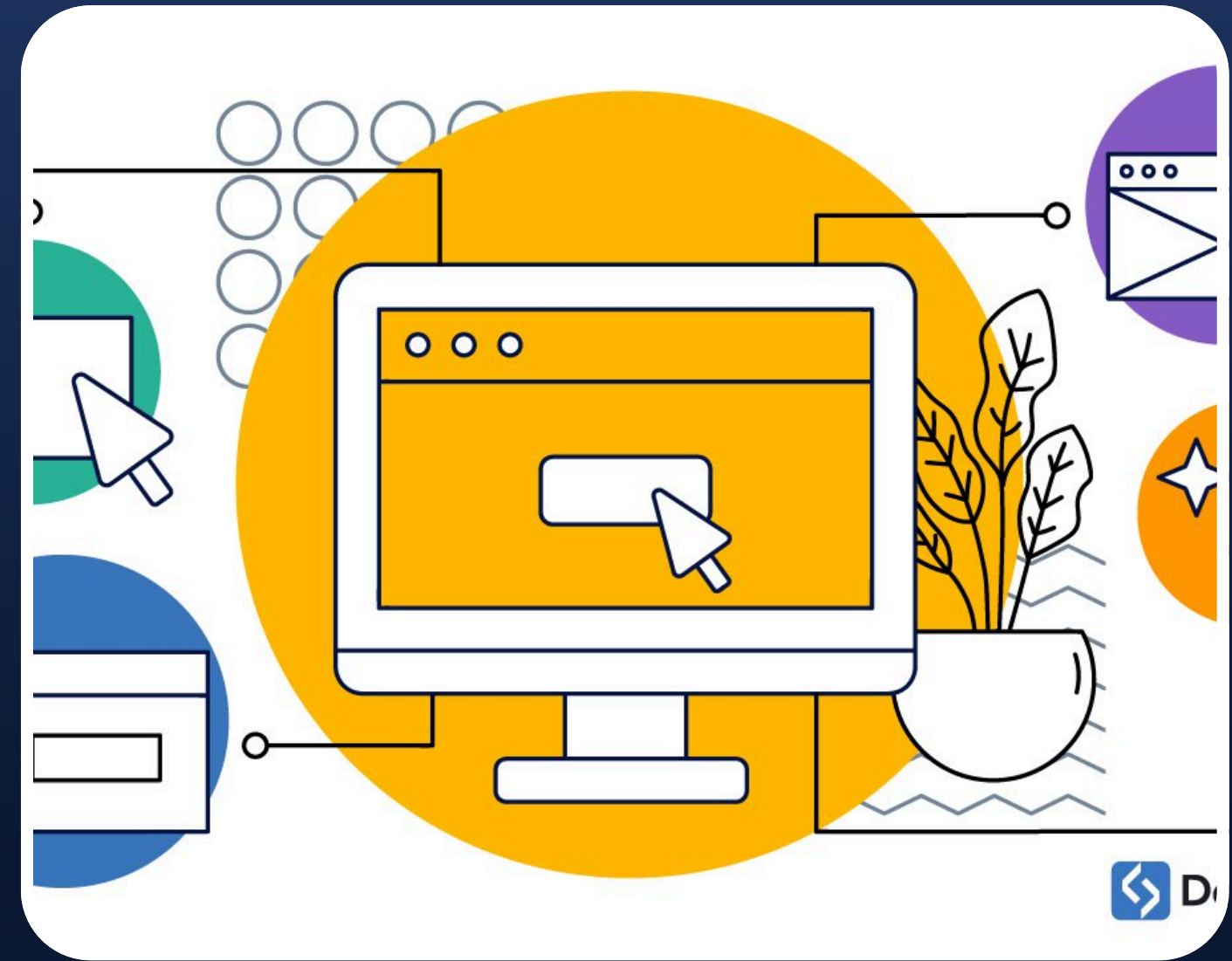


Considers real poker factors:

- Hand strength
- Pot odds
- Calling cost
- Player position
- Stack size



Makes AI decisions strategic, not greedy



Balanced Heuristic Evaluation – Significance

Group-07 (Saba , Ema , Sumaya , Fahim)



Designed for games with:

- Opponents (MIN nodes)
- AI decisions (MAX nodes)
- Random events (CHANCE nodes)



Searches future states to choose best move

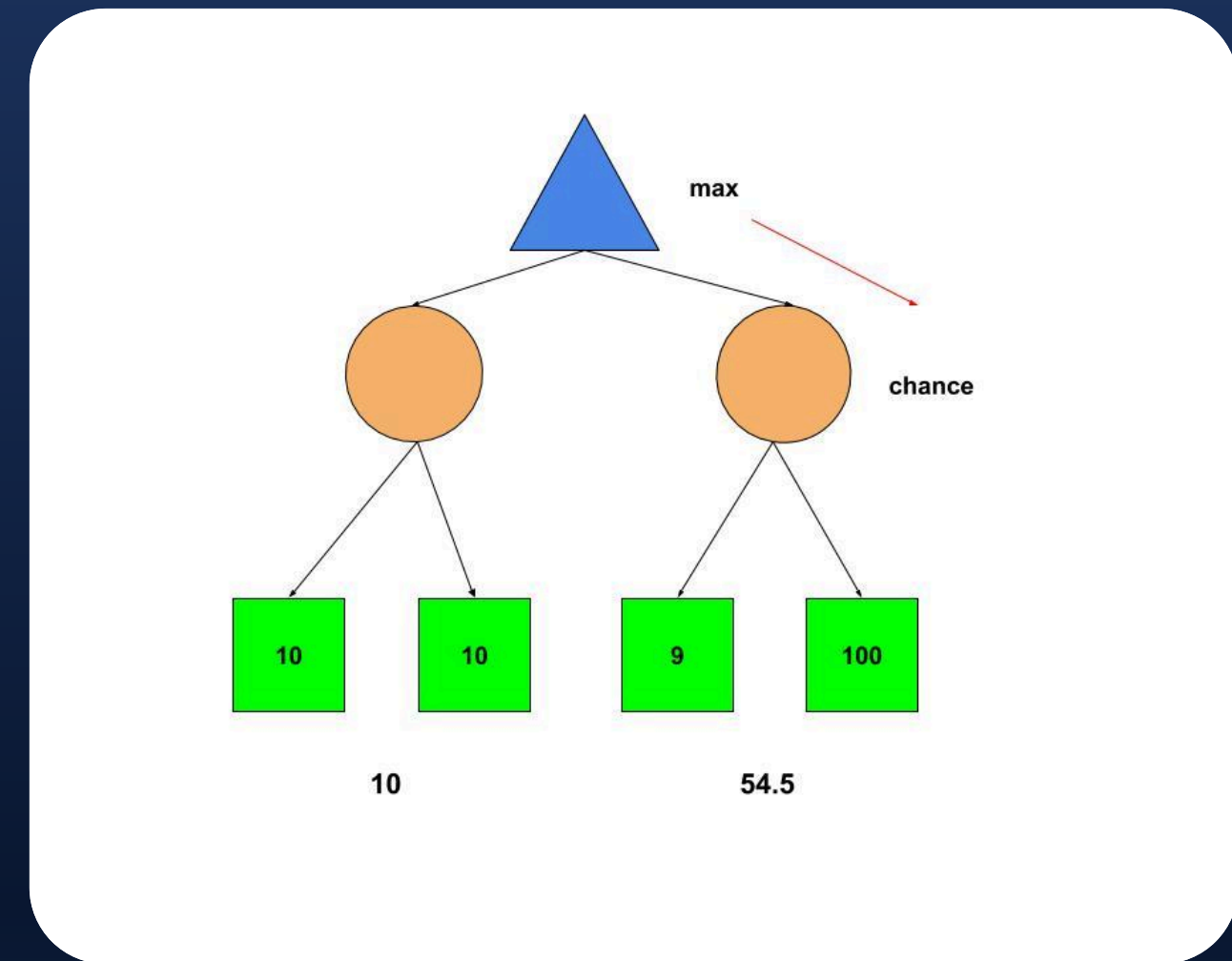


Models real poker flow:

- Betting rounds
- Card dealing
- Opponent actions



Handles uncertainty better than minimax



Expectiminimax Algorithm – Significance

Results

TABLE I
DETAILED OPPONENT STATISTICS

Opp.	Strat.	W(%)	Pft	ROI	H	LW	LL
Call St.	Always calls	66.7	990	99.0	6	950	40
Agg.	Constant raises	54.8	980	98.0	31	900	1080
Tight	Strong hands only	47.3	990	99.0	74	1080	900
Random	Random actions	51.3	980	98.0	39	1920	1080
Passive	Rarely raises	42.3	-1000	-100	26	360	1080

✓ AI Vs Deterministic

✓ AI Vs AI

Ai vs Deterministic simulation
er overall ai win rate: 49.4%
total hands played 176

Depth	Hands	Wins (A1,A2)	Chips (A1,A2)	Winner
(2,2)	5	(2,3)	(0,2000)	A2
(3,3)	3	(1,2)	(0,1920)	A2
(2,3)	5	(1,4)	(220,1780)	A2
(3,2)	5	(1,4)	(170,1830)	A2

Project Significance (Overall)

- ✓ Solves imperfect information problem
- ✓ Mimics real poker decision-making
- ✓ Combines probability + strategy + prediction
- ✓ Produces stable and explainable decisions
- ✓ Suitable for research and AI game modeling



Future Work



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Conclusion

- ✓ Built a strategic Texas Hold'em AI
- ✓ Successfully handled uncertainty and randomness
- ✓ Integrated Monte Carlo, heuristic, and expectiminimax
- ✓ AI makes rational and balanced decisions
- ✓ Demonstrates effective AI decision-making in complex games

